

CS-E4715 Supervised Machine Learning

Lecture 3: Learning with infinite hypothesis classes

Recall: PAC learnability

- A class C is PAC-learnable, if there exist an algorithm $\mathcal{A}$ that given a training sample S outputs a hypothesis h_S that has generalization error satisfying

$$Pr(R(h_S) \leq \epsilon) \geq 1 - \delta$$

- for any distribution D , for arbitrary $\epsilon, \delta > 0$ and sample size $m = |S|$ that grows at polynomially in $1/\epsilon, 1/\delta$

Recall: PAC learning of a finite hypothesis class

- Sample complexity bound relying on the size of the hypothesis class (Mohri et al, 2018): $Pr(R(h_s) \leq \epsilon) \geq 1 - \delta$ if

$$m \geq \frac{1}{\epsilon} (\log(|\mathcal{H}|) + \log(\frac{1}{\delta}))$$

- An equivalent generalization error bound:

$$R(h) \leq \frac{1}{m} (\log(|\mathcal{H}|) + \log(\frac{1}{\delta}))$$

- Holds for any finite hypothesis class assuming there is a consistent hypothesis, one with zero empirical risk
- Extra term compared to the rectangle learning example is the term $\frac{1}{\epsilon} (\log(|\mathcal{H}|))$
- The more hypotheses there are in $\mathcal{H}$, the more training examples are needed

Learning with infinite hypothesis classes

- The size of the hypothesis class is a useful measure of complexity for **finite** hypothesis classes (e.g. boolean formulae)
- However, most classifiers used in practise rely on infinite hypothesis classes, e.g.
 - $\mathcal{H}$ = axis-aligned rectangles in $\mathbb{R}^2$ (the example last lecture)
 - $\mathcal{H}$ = hyperplanes in $\mathbb{R}^d$ (e.g. Support vector machines)
 - $\mathcal{H}$ = neural networks with continuous input variables
- Need better tools to analyze these cases

Vapnik-Chervonenkis dimension

- VC dimension can be understood as measuring the capacity of a hypothesis class to adapt to different concepts
- It can be understood through the following thought experiment:
 - Pick a fixed hypothesis class $\mathcal{H}$, e.g. axis-aligned rectangles in R^2
 - Let us enumerate all possible labelings of a training set of size m : $\mathcal{Y}^m = \{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_{2^m}\}$, where $\mathbf{y}_j = (y_{j1}, \dots, y_{jm})$, and $y_{ij} \in \{0, 1\}$ is the label of i 'th example in the j 'th labeling
 - We are allowed to freely choose a distribution D generating the inputs and to generate the input data $x_1, \dots, x_m$
 - $VCdim(\mathcal{H}) =$ size of the **largest training set** that we can find a consistent classifier for **all labelings** in $\mathcal{Y}^m$
- Intuitively:
 - low $VCdim \implies$ easy to learn, low sample complexity
 - high $VCdim \implies$ hard to learn, high sample complexity
 - infinite $VCdim \implies$ cannot learn in PAC framework

Shattering

- The underlying concept in VC dimension is **shattering**
- Given a set of points $S = \{x_1, \dots, x_m\}$ and a fixed class of functions $\mathcal{H}$
- $\mathcal{H}$ is said to **shatter** S if for any possible partition of S into positive S_+ and negative subset S_- we can find a hypothesis for which $h(x) = 1$ if and only if $x \in S_+$

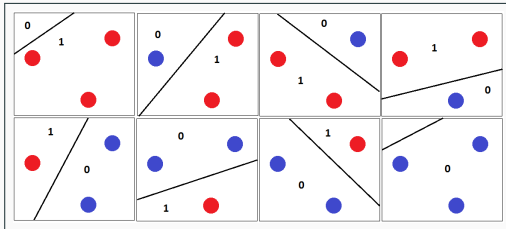


Figure source:

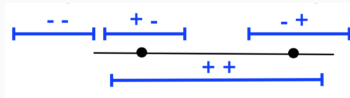
<https://datascience.stackexchange.com>

How to show that $VCdim(\mathcal{H}) = d$

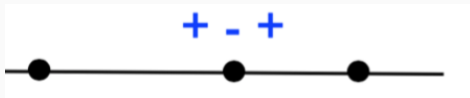
- How to show that $VCdim(\mathcal{H}) = d$ for a hypothesis class
- We need to show two facts:
 1. There **exists a set of inputs** of size d that can be shattered by hypothesis in $\mathcal{H}$ (i.e. we can pick the set of inputs any way we like):
 $VCdim(\mathcal{H}) \geq d$
 2. There does not exist **any set of inputs** of size $d + 1$ that can be shattered (i.e. need to show a general property): $VCdim(\mathcal{H}) < d + 1$

Example: intervals on a real line

- Let the hypothesis class be intervals in $\mathbb{R}$
- Each hypothesis is defined by two parameters $b_h, e_h \in \mathbb{R}$: the beginning and end of the interval, $h(x) = \mathbf{1}_{b_h \leq x \leq e_h}$
- We can shatter any set of two points by changing the end points of the interval:



- We cannot shatter a three point set, as the middle point cannot be excluded while the left-hand and right-hand side points are included



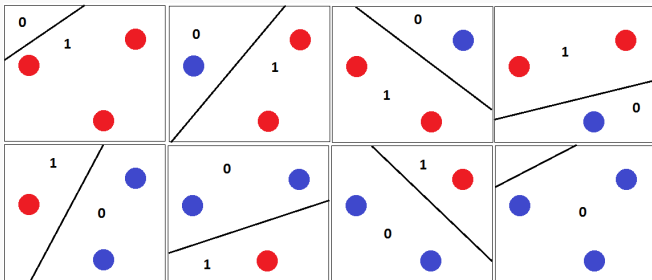
We conclude that VC dimension for real intervals = 2

Linear classifiers in $\mathbb{R}^2$

- Consider affine halfspace classifiers

$$h_{\mathbf{w},b}(\mathbf{x}) = \mathbf{1}_{\mathbf{w}^\top \mathbf{x} + b \geq 0}, \quad \mathbf{w} \in \mathbb{R}^2, \quad b \in \mathbb{R}.$$

- Any three non-collinear points can be shattered by this class.

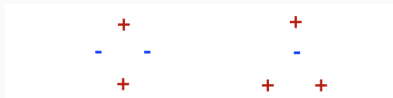


- We conclude that VC dimension is ≥ 3

Linear classifiers in $\mathbb{R}^2$

No set of four distinct points can be shattered:

- **Three points are collinear:** label the two outer points 1 and the middle point 0. A halfspace containing both outer points also contains the segment between them.
- **No three are collinear:** there are two cases:
 - A convex quadrilateral: alternate labels around its boundary. The segments joining each same-label pair intersect, so no line separates the classes.
 - A triangle with one point inside: label the vertices 1 and the interior point 0. A halfspace containing the vertices contains their convex hull.
- Together with the three-point lower bound, $VCdim(\mathcal{H}) = 3$.



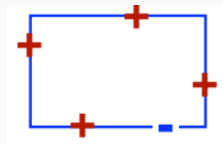
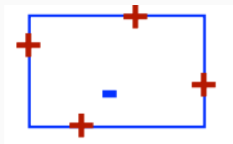
VC-dimension of axis-aligned rectangles

- With axis aligned rectangles we can shatter a set of four points (picture shows 4 of the 16 configurations)
- This implies $VCdim(\mathcal{H}) \geq 4$



VC-dimension of axis-aligned rectangles

- For five distinct points, consider the minimum bounding box of the points
- There are two possible configurations:
 1. There are one or more points in the interior of the box: then one cannot include the points on the boundary and exclude the points in the interior
 2. At least one of the edges contains two points: in this case we can pick either of the two points and verify that this point cannot be excluded while all the other points are included
- Thus by the two examples we have established that $VCdim(\mathcal{H}) = 4$



Vapnik-Chervonenkis dimension formally

- Formally $VCdim(\mathcal{H})$ is defined through the growth function

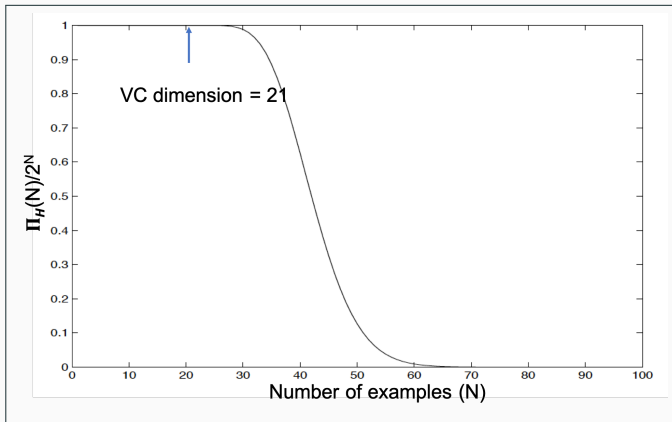
$$\Pi_{\mathcal{H}}(m) = \max_{\{x_1, \dots, x_m\} \subset X} |\{(h(x_1), \dots, h(x_m)) : h \in \mathcal{H}\}|$$

- The growth function gives the maximum number of unique labelings the hypothesis class $\mathcal{H}$ can provide for an arbitrary set of input points
- The maximum of the growth function is 2^m for a set of m examples
- Vapnik-Chervonenkis dimension is then

$$VCdim(\mathcal{H}) = \max_m \{m | \Pi_{\mathcal{H}}(m) = 2^m\}$$

Visualization

- The ratio of the growth function $\Pi_{\mathcal{H}}(m)$ to the maximum number of labelings of a set of size m is shown
- Hypothesis class is 20-dimensional hyperplanes (VC dimension = 21)



VC dimension of finite hypothesis classes

- Any finite hypothesis class has VC dimension $VCdim(\mathcal{H}) \leq \log_2 |\mathcal{H}|$
- To see this:
 - Consider a set of m examples $S = \{x_1, \dots, x_m\}$
 - This set can be labeled 2^m different ways, by choosing the labels $y_i \in \{0, 1\}$ independently
 - Each hypothesis in $h \in \mathcal{H}$ fixes one labeling, a length- m binary vector $\mathbf{y}(h, S) = (h(x_1), \dots, h(x_m))$
 - All hypotheses in $\mathcal{H}$ together can provide at most $|\mathcal{H}|$ different labelings in total (different vectors $\mathbf{y}(h, S), h \in \mathcal{H}$)
 - If $|\mathcal{H}| < 2^m$ we cannot shatter $S \implies$ we cannot shatter a set of size $m > \log_2 |\mathcal{H}|$

VC dimension: Further examples

Examples of classes with a finite VC dimension:

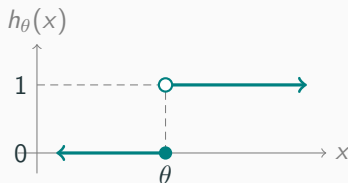
- convex d -polygons in $\mathbb{R}^2$: $VCdim = 2d + 1$ (e.g. for general, not restricted to axis-aligned, rectangles $VCdim = 5$)
- hyperplanes in $\mathbb{R}^d$: $VCdim = d + 1$ - (e.g. single neural unit, linear SVM)
- neural networks: $VCdim = |E| \log |E|$ where E is the set of edges in the networks (for *sign* activation function)
- boolean monomials of d variables: $VCdim = d$
- arbitrary boolean formulae of d variables: $VCdim = 2^d$

Half-time poll: VC dimension of threshold functions in $\mathbb{R}$

Consider a hypothesis class $\mathcal{H} = \{h_\theta\}$ of threshold functions

$h_\theta : \mathbb{R} \mapsto \{0, 1\}$, $\theta \in \mathbb{R}$:

$$h_\theta(x) = \begin{cases} 1 & \text{if } x > \theta \\ 0 & \text{otherwise} \end{cases}$$



What is the VC dimension of this hypothesis class?

1. $VCdim = 1$
2. $VCdim = 2$
3. $VCdim = \infty$

Answer to the poll in Mycourses by 11:15: Go to Lectures page and scroll down to "Lecture 3 poll":

Answers are anonymous and do not affect grading of the course.

Half-time poll: Answer

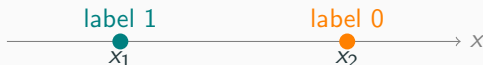
$$\text{Answer: } VCdim(\mathcal{H}) = 1 \quad h_{\theta}(x) = \mathbf{1}_{x > \theta}$$

One point can be shattered. For any point x_1 , both labels are possible:

$$\theta < x_1 \implies h_{\theta}(x_1) = 1, \quad \theta \geq x_1 \implies h_{\theta}(x_1) = 0.$$

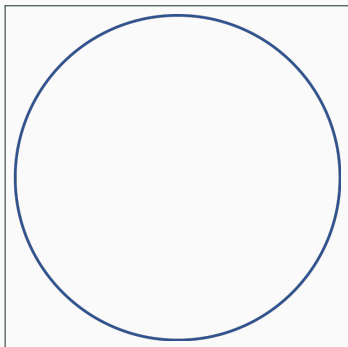
No pair of points can be shattered. For any $x_1 < x_2$, the labeling $(1, 0)$ is impossible:

$$h_{\theta}(x_1) = 1 \implies \theta < x_1 < x_2 \implies h_{\theta}(x_2) = 1.$$



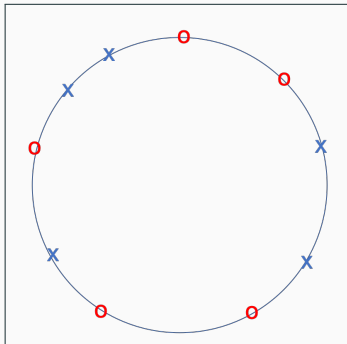
Convex polygons have VC dimension $= \infty$

- Let our hypothesis class be convex polygons in $\mathbb{R}^2$ without restriction of number of vertices d
- Let us draw an arbitrary circle on $\mathbb{R}^2$ - the distribution D will be concentrated on the circumference of the circle
 - This is a difficult distribution for learning polygons - we choose it on purpose



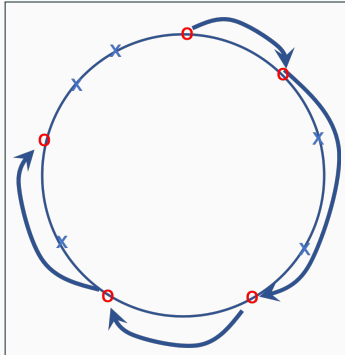
Convex polygons have VC dimension = ∞

- Let us consider a set of m points with arbitrary binary labels
- For any m , let us position m points on the circumference of the circle
 - simulating drawing the inputs from the distribution D



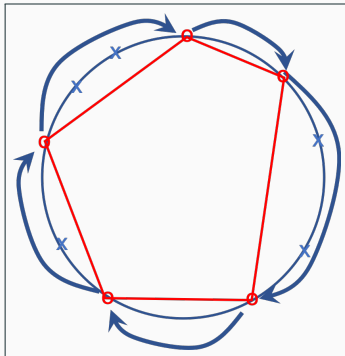
Convex polygons have VC dimension = ∞

- Start from an arbitrary positive point (red circles)
- Traverse the circumference clockwise skipping all negative points and stopping at positive points



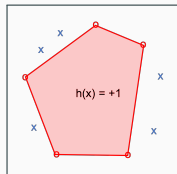
Convex polygons have VC dimension = ∞

- Connect adjacent positive points with an edge
- This forms a p -polygon inside the circle, where p is the number of positive data points



Convex polygons have VC dimension $= \infty$

- Define $h(x) = +1$ for points inside the polygon and $h(x) = 0$ outside
- Each of the 2^m labelings of m examples gives us a p -polygon that includes the p positive points in that labeling and excludes the negative points $\implies$ we can shatter a set of size m : $VCdim(\mathcal{H}) \geq m$
- Since m was arbitrary, we can grow it without limit $VCdim(\mathcal{H}) = \infty$



Generalization bound based on the VC-dimension

- (Mohri, 2018) Let $\mathcal{H}$ be a family of functions taking values in $\{-1, +1\}$ with VC-dimension d . Then for any $\delta > 0$, with probability at least $1 - \delta$ the following holds for all $h \in \mathcal{H}$:

$$R(h) \leq \hat{R}(h) + \sqrt{\frac{2 \log(em/d)}{m/d}} + \sqrt{\frac{\log(1/\delta)}{2m}}$$

- $e \approx 2.71828$ is the base of the natural logarithm
- The bound reveals that the critical quantity is m/d , i.e. the number of examples divided by the VC-dimension
- Manifestation of the Occam's razor principle: to justify an increase in the complexity, we need reciprocally more data

Rademacher complexity

Experiment: how well does your hypothesis class fit noise?

- Consider a set of training examples $S_0 = \{(x_i, y_i)\}_{i=1}^m$
- Generate M new datasets $S_1, \dots, S_M$ from S_0 by randomly drawing a new label $\sigma \in \mathcal{Y}$ for each training example in S_0

$$S_k = \{(x_i, \sigma_{ik})\}_{i=1}^m$$

- Train a classifier h_k minimizing the empirical risk on training set S_k , record its empirical risk

$$\hat{R}(h_k) = \frac{1}{m} \sum_{i=1}^m \mathbf{1}_{h_k(x_i) \neq \sigma_{ik}}$$

- Compute the average empirical risk over all datasets:
 $\bar{\epsilon} = \frac{1}{M} \sum_{k=1}^M \hat{R}(h_k)$

Experiment: how well does your hypothesis class fit noise?

- Observe the quantity

$$\hat{\mathcal{R}} = \frac{1}{2} - \bar{\epsilon}$$

- We have $\hat{\mathcal{R}} = 0$ when $\bar{\epsilon} = 0.5$, that is when the predictions correspond to random coin flips (0.5 probability to predict either class)
- We have $\hat{\mathcal{R}} = 0.5$ when $\bar{\epsilon} = 0$, that is when all hypotheses $h_i, i = 1, \dots, M$ have zero empirical error (perfect fit to noise, not good!)
- Intuitively we would like our hypothesis
 - to be able to separate noise from signal - to have low $\hat{\mathcal{R}}$
 - have low empirical error on real data - otherwise impossible to obtain low generalization error

Rademacher complexity

- Rademacher complexity defines complexity as the capacity of hypothesis class to fit random noise
- For binary classification with labels $\mathcal{Y} = \{-1, +1\}$ empirical Rademacher complexity can be defined as

$$\hat{\mathcal{R}}_S(\mathcal{H}) = \frac{1}{2} E_{\sigma} \left(\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \sigma_i h(\mathbf{x}_i) \right)$$

- $\sigma_i \in \{-1, +1\}$ are Rademacher random variables, drawn independently from uniform distribution (i.e. $Pr\{\sigma = 1\} = 0.5$)
- Expression inside the expectation takes the highest correlation over all hypothesis in $h \in \mathcal{H}$ between the random true labels σ_i and predicted label $h(\mathbf{x}_i)$

$$\hat{\mathcal{R}}_S(\mathcal{H}) = \frac{1}{2} E_{\sigma} \left(\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \sigma_i h(\mathbf{x}_i) \right)$$

- Let us rewrite $\hat{\mathcal{R}}_S(\mathcal{H})$ in terms of empirical error
- Note that with labels $\mathcal{Y} = \{+1, -1\}$,

$$\sigma_i h(\mathbf{x}_i) = \begin{cases} 1 & \text{if } \sigma_i = h(\mathbf{x}_i) \\ -1 & \text{if } \sigma_i \neq h(\mathbf{x}_i) \end{cases}$$

- Thus

$$\begin{aligned} \frac{1}{m} \sum_{i=1}^m \sigma_i h(\mathbf{x}_i) &= \frac{1}{m} \left(\sum_i \mathbf{1}_{\{h(\mathbf{x}_i) = \sigma_i\}} - \sum_i \mathbf{1}_{\{h(\mathbf{x}_i) \neq \sigma_i\}} \right) \\ &= \frac{1}{m} (m - 2 \sum_i \mathbf{1}_{\{h(\mathbf{x}_i) \neq \sigma_i\}}) = 1 - 2\hat{\epsilon}(h) \end{aligned}$$

Rademacher complexity

- Plug in

$$\begin{aligned}\hat{\mathcal{R}}_S(\mathcal{H}) &= \frac{1}{2} E_{\sigma} \left(\sup_{h \in \mathcal{H}} (1 - 2\hat{\epsilon}(h)) \right) \\ &= \frac{1}{2} (1 - 2E_{\sigma} \inf_{h \in \mathcal{H}} \hat{\epsilon}(h)) = \frac{1}{2} - E_{\sigma} \inf_{h \in \mathcal{H}} \hat{\epsilon}(h)\end{aligned}$$

- Now we have expressed the empirical Rademacher complexity in terms of expected empirical error of classifying randomly labeled data
- But how does the Rademacher complexity help in model selection?
 - We need to relate it to generalization error

Generalization bound with Rademacher complexity

(Mohri et al. 2018): For any $\delta > 0$, with probability at least $1 - \delta$ over a sample drawn from an unknown distribution D , for any $h \in \mathcal{H}$ we have:

$$R(h) \leq \hat{R}_S(h) + 2\hat{\mathcal{R}}_S(\mathcal{H}) + 3\sqrt{\frac{\log \frac{2}{\delta}}{2m}}$$

The bound is composed of the sum of :

- The empirical risk of h on the training data S (with the original labels): $\hat{R}_S(h)$
- Twice the empirical Rademacher complexity (with our normalization): $2\hat{\mathcal{R}}_S(\mathcal{H})$
- A term that tends to zero as a function of size of the training data as $O(1/\sqrt{m})$ assuming constant δ .

Optional proof: from classifiers to losses

Not examinable — optional proof.

Let $S = (z_1, \dots, z_m) \sim D^m$, where $z_i = (x_i, y_i)$, and $h(x), y \in \{-1, +1\}$.

Define the loss class

$$\mathcal{G} = \{g_h : (x, y) \mapsto (1 - yh(x))/2 : h \in \mathcal{H}\}.$$

Every g_h takes values in $\{0, 1\} \subset [0, 1]$. Write its standard empirical Rademacher complexity as

$$\hat{r}_S(\mathcal{G}) = \mathbb{E}_\sigma \sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^m \sigma_i g(z_i).$$

Since $\mathbb{E}_\sigma \sum_i \sigma_i = 0$ and $(-\sigma_i y_i)_i$ has the same distribution as $(\sigma_i)_i$, for fixed S ,

$$\hat{r}_S(\mathcal{G}) = \frac{1}{2} \mathbb{E}_\sigma \sup_{h \in \mathcal{H}} \frac{1}{m} \sum_i \sigma_i h(x_i) = \hat{\mathcal{R}}_S(\mathcal{H}).$$

The last equality uses **this lecture's normalization**.

Optional proof: symmetrization

Not examinable — optional proof.

Define the largest generalization gap and expected complexity:

$$\Phi(S) = \sup_{g \in \mathcal{G}} \left(\mathbb{E}_D g - \frac{1}{m} \sum_i g(z_i) \right), \quad r_m = \mathbb{E}_S \hat{r}_S(\mathcal{G}).$$

Introduce an independent sample $S' = (z'_1, \dots, z'_m) \sim D^m$. Moving the expectation inside the supremum gives an upper bound:

$$\begin{aligned} \mathbb{E}_S \Phi(S) &\leq \mathbb{E}_{S, S'} \sup_g \frac{1}{m} \sum_i (g(z'_i) - g(z_i)) \\ &= \mathbb{E}_{S, S', \sigma} \sup_g \frac{1}{m} \sum_i \sigma_i (g(z'_i) - g(z_i)) \\ &\leq \mathbb{E}_{S', \sigma} \sup_g \frac{1}{m} \sum_i \sigma_i g(z'_i) + \mathbb{E}_{S, \sigma} \sup_g \frac{1}{m} \sum_i (-\sigma_i) g(z_i) \\ &= 2r_m. \end{aligned}$$

The equality holds because each pair (z_i, z'_i) is exchangeable: randomly swapping its entries does not change its distribution.

Optional proof: a high-probability bound

Not examinable — optional proof.

Changing one observation changes either $\Phi(S)$ or $\hat{r}_S(\mathcal{G})$ by at most $1/m$, since $g \in [0, 1]$. For $0 < \delta < 1$, set $t = \sqrt{\log(2/\delta)/(2m)}$.

McDiarmid's bounded-differences inequality gives, respectively,

$$\Phi(S) \leq \mathbb{E}\Phi(S) + t, \quad r_m \leq \hat{r}_S(\mathcal{G}) + t,$$

each with failure probability at most $\delta/2$. By the union bound, both hold with probability at least $1 - \delta$:

$$\Phi(S) \leq 2r_m + t \leq 2\hat{r}_S(\mathcal{G}) + 3t.$$

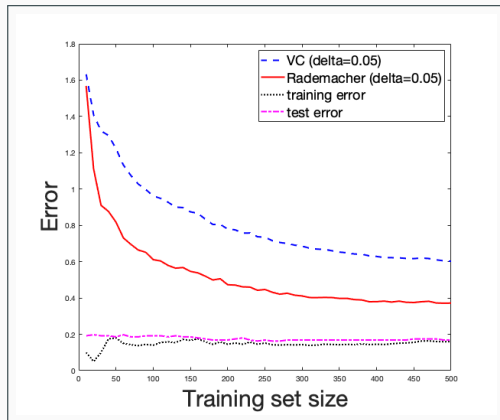
Thus, simultaneously for every $h \in \mathcal{H}$,

$$R(h) \leq \hat{R}_S(h) + 2\hat{\mathcal{R}}_S(\mathcal{H}) + 3\sqrt{\frac{\log(2/\delta)}{2m}}.$$

Proof outline following Mohri et al. (2018); standard measurability assumptions apply.

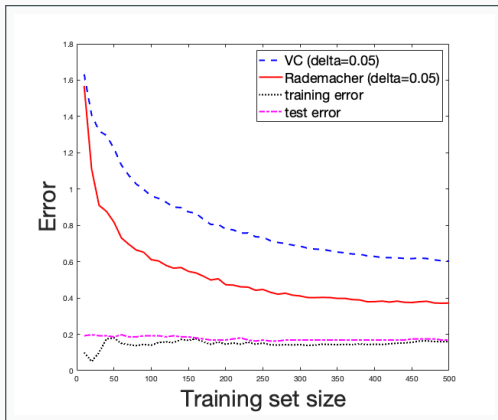
Example: Rademacher and VC bounds on a real dataset

- Prediction of protein subcellular localization
- 10-500 training examples, 172 test examples
- Comparing Rademacher and VC bounds using $\delta = 0.05$
- Training and test error also shown



Example: Rademacher and VC bounds on a real dataset

- Rademacher bound is sharper than the VC bound
- VC bound is not yet informative with 500 examples (> 0.5) using ($\delta = 0.05$)
- The gap between the mean of the error distribution ($\approx$ test error) and the 0.05 probability tail (VC and Rademacher bounds) is evident (and expected)



Note the differences between Rademacher complexity and VC dimension

- VC dimension is independent of any training sample or distribution generating the data: it measures the worst-case where the data is generated in a bad way for the learner
- Rademacher complexity depends on the training sample thus is dependent on the data generating distribution
- VC dimension focuses the extreme case of realizing all labelings of the data
- Rademacher complexity measures smoothly the ability to realize random labelings

- Generalization bounds based on Rademacher Complexity are applicable to any binary classifiers (SVM, neural network, decision tree)
- It motivates state of the art learning algorithms such as support vector machines
- But computing it might be hard, if we need to train a large number of classifiers
- Vapnik-Chervonenkis dimension (VCdim) is an alternative that is usually easier to derive analytically

Summary: Statistical learning theory

- Statistical learning theory focuses in analyzing the generalization ability of learning algorithms
- Probably Approximately Correct framework is the most studied theoretical framework, asking for bounding the generalization error (ϵ) with high probability ($1 - \delta$), with arbitrary level of error $\epsilon > 0$ and confidence $\delta > 0$
- Vapnik-Chervonenkis dimension lets us study learnability infinite hypothesis classes through the concept of shattering
- Rademacher complexity is a practical alternative to VC dimension, giving typically sharper bounds (but requires a lot of simulations to be run)